Rocket Flight Simulator and Mission Analysis Tool

Mission Report

Mission Summary

This report summarizes a one-dimensional vertical launch using variable thrust, propellant mass depletion, altitude-dependent atmospheric density, and velocity-opposed aerodynamic drag.

Rocket Parameters

Parameter	Value
Dry Mass	3.000
Fuel Mass	100.000
Thrust	300.000
Burn Time	8.000
Drag Coefficient	0.450
Cross Sectional Area	0.020

Atmospheric Model

Density follows a simplified International Standard Atmosphere: 1.225 kg/m^3 at sea level, cooling through the troposphere, then an isothermal lower-stratosphere approximation above 11 km.

Flight Phase Timeline

Phase	Start (s)	End (s)	Duration (s)
PRE-LAUNCH	0.000	5.700	5.700
LIFTOFF	5.800	5.800	0.000
POWERED_ASCENT	5.900	7.900	2.000
COASTING	8.000	9.400	1.400
APOGEE	9.500	9.500	0.000
DESCENT	9.600	11.600	2.000
LANDING	11.700	11.700	0.000

Performance Metrics

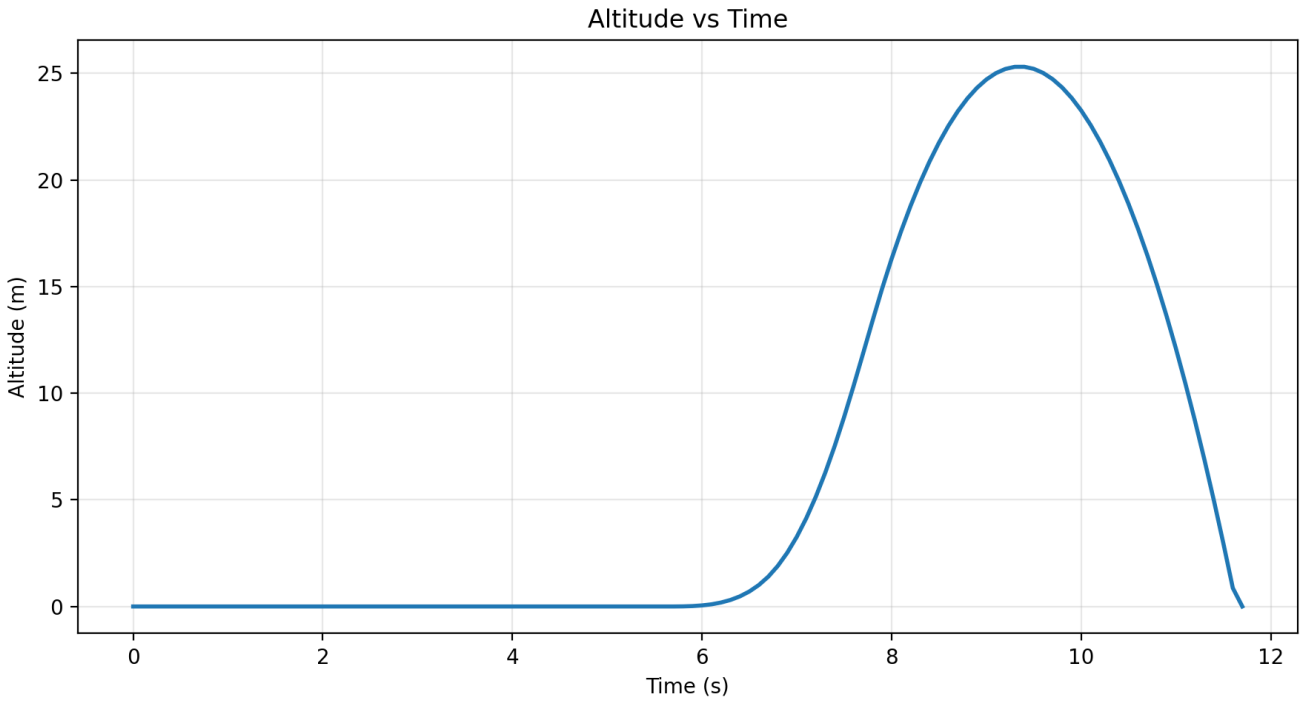
Metric	Value
Maximum Altitude M	25.306

Maximum Velocity Mps	21.897
Maximum Acceleration Mps2	13.051
Burnout Time S	7.900
Apogee Time S	9.400
Total Flight Duration S	11.700
Landing Velocity Mps	-21.897
Flight Duration S	11.700

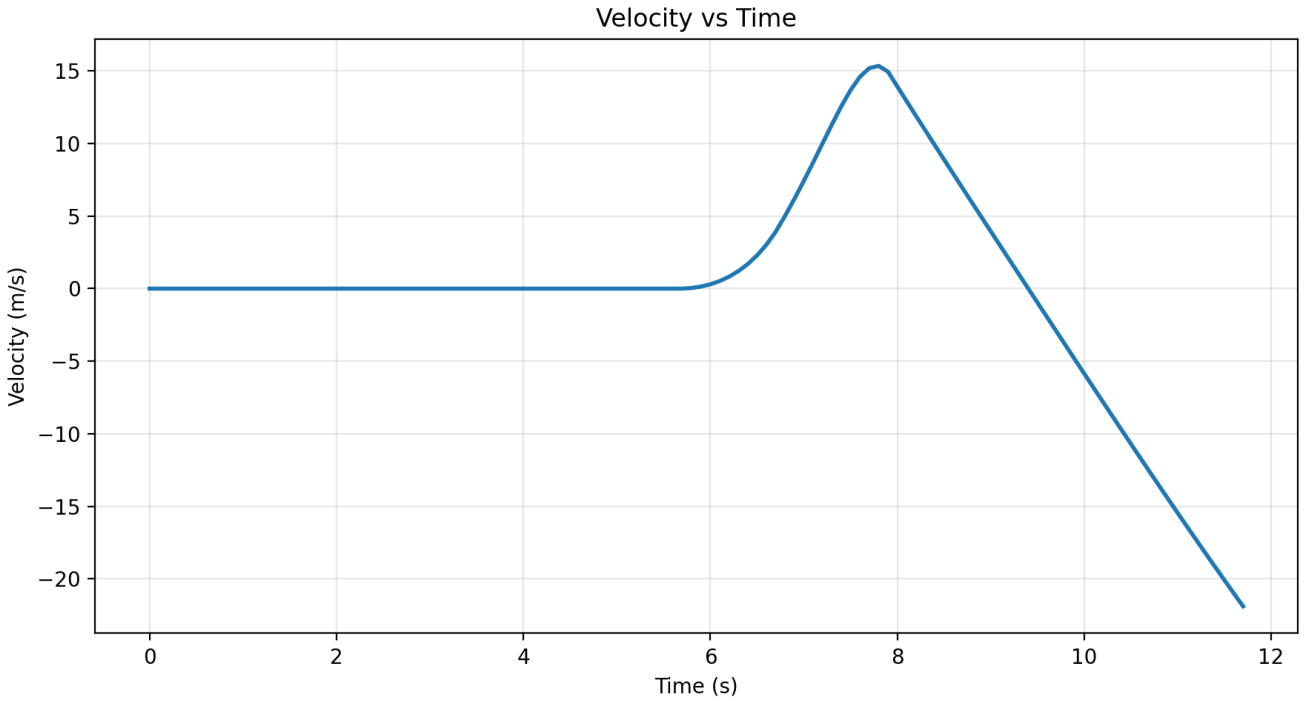
Engineering Conclusions

The upgraded simulation captures the dominant first-order effects for vertical flight analysis: decreasing density reduces drag at altitude, thrust tail-off shapes burnout acceleration, and mass depletion improves the predicted acceleration history.

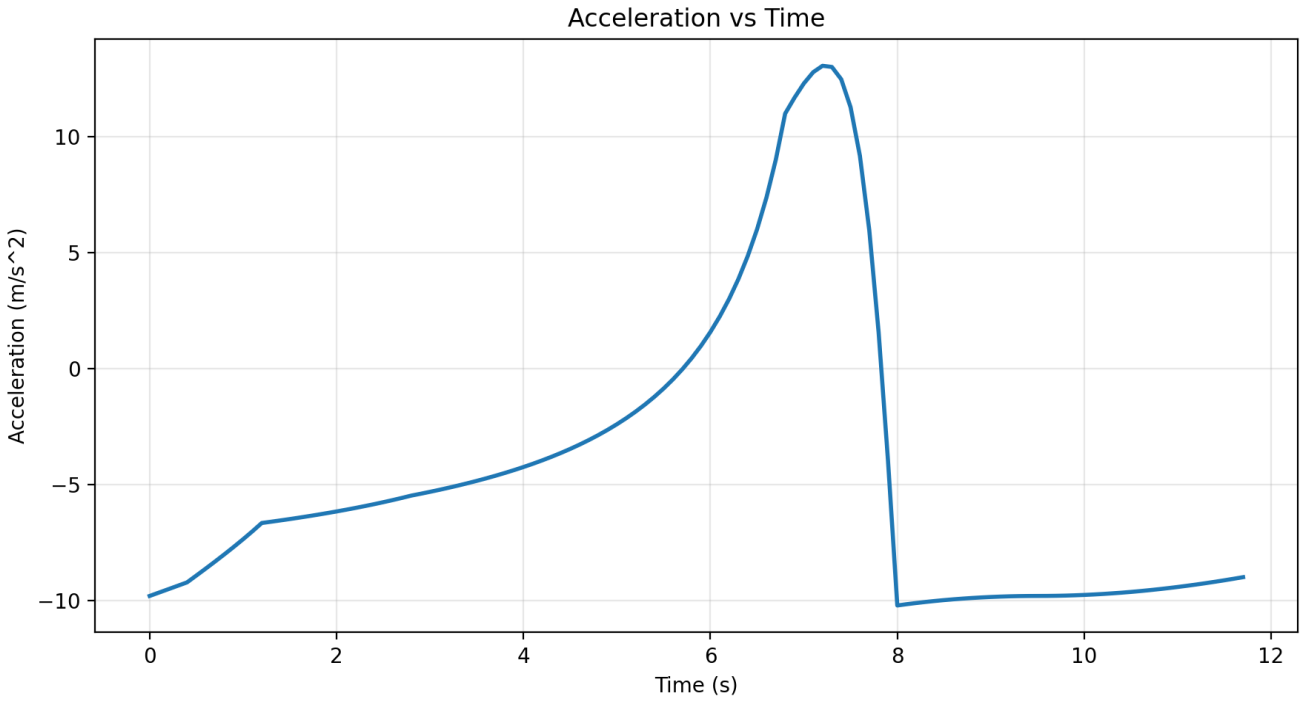
Altitude Plot



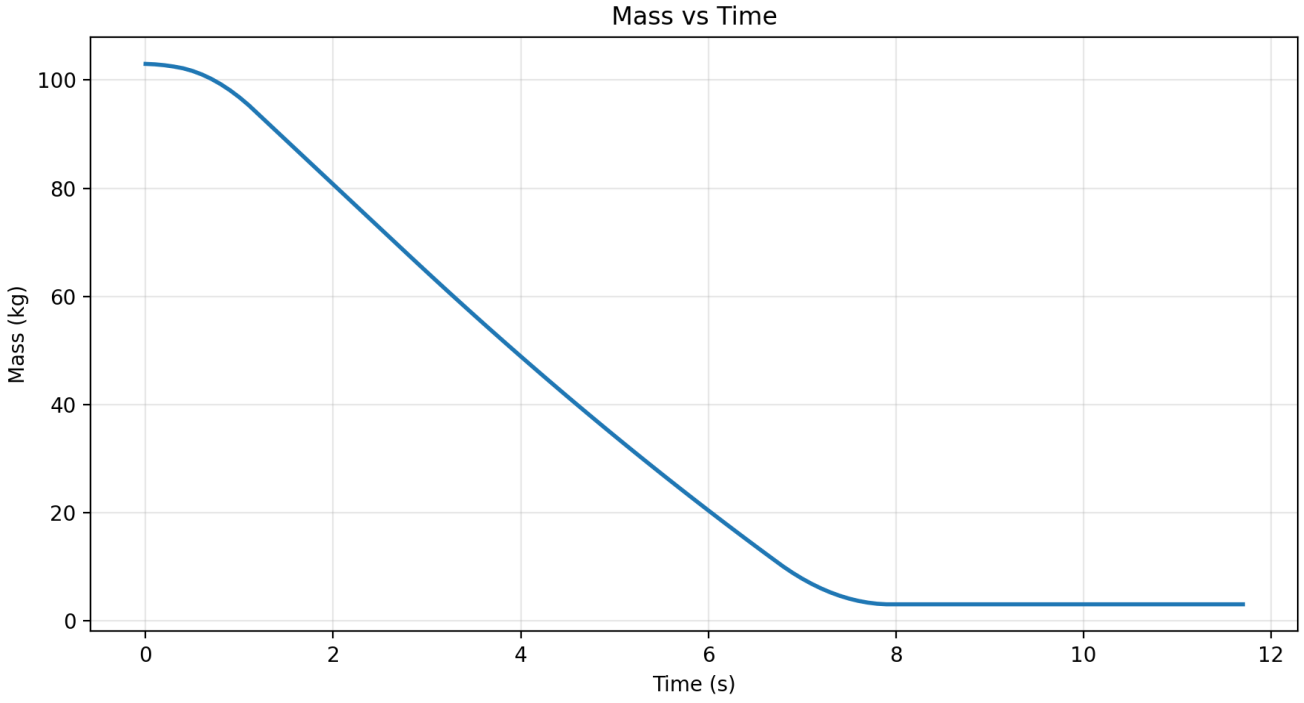
Velocity Plot



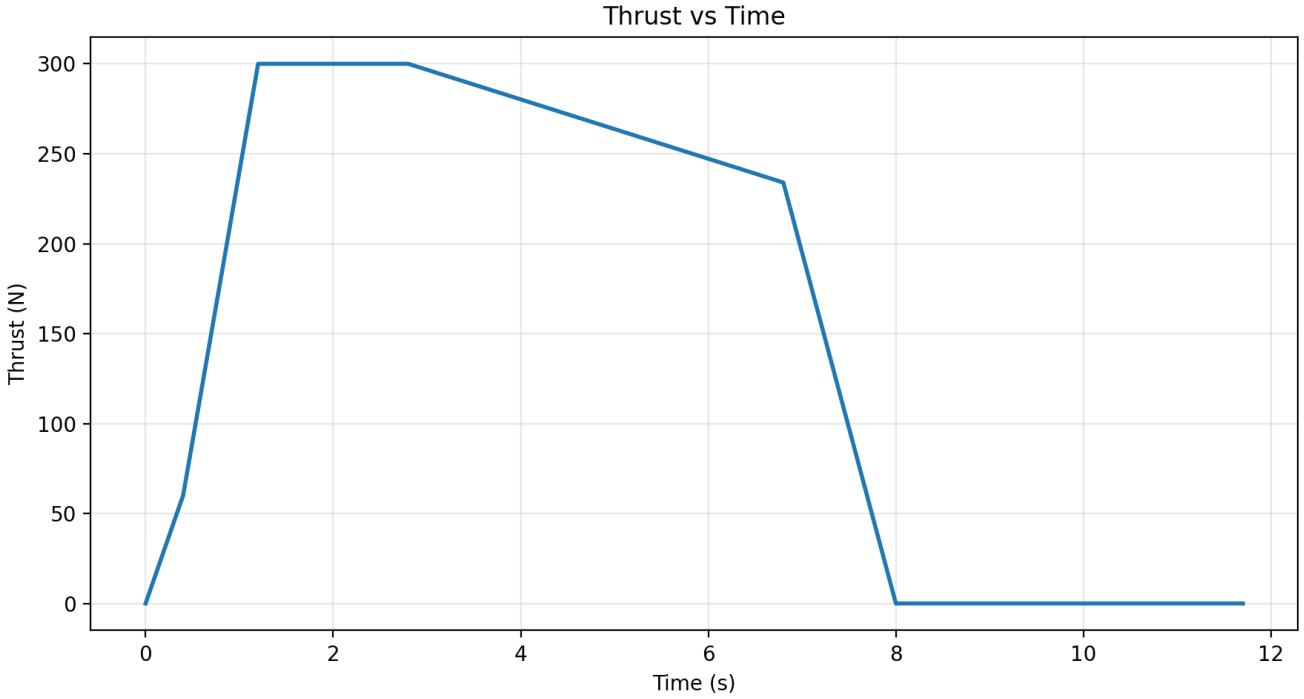
Acceleration Plot



Mass Plot



Thrust Plot


Thrust vs Time
Thrust (N)
Time (s)
300
250
200
150
100
50
0
0
2
4
6
8
10
12